DeepNeoAG Learning Through Controlled Experiments

Computational Biology
Course investigation

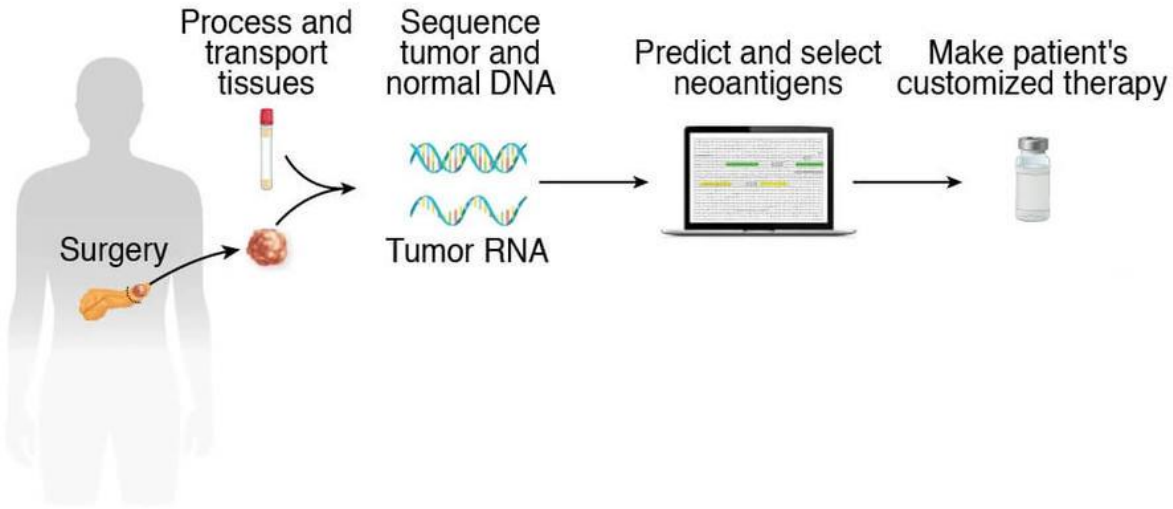
Complete experimental run
25 October 2026

Composition + LR / One-hot MCNN / ESM pooled LR / ESM CNN / ESM MCNN

Why predict neoantigen-like peptides?

Sequence prediction covers one part of a wider biological process

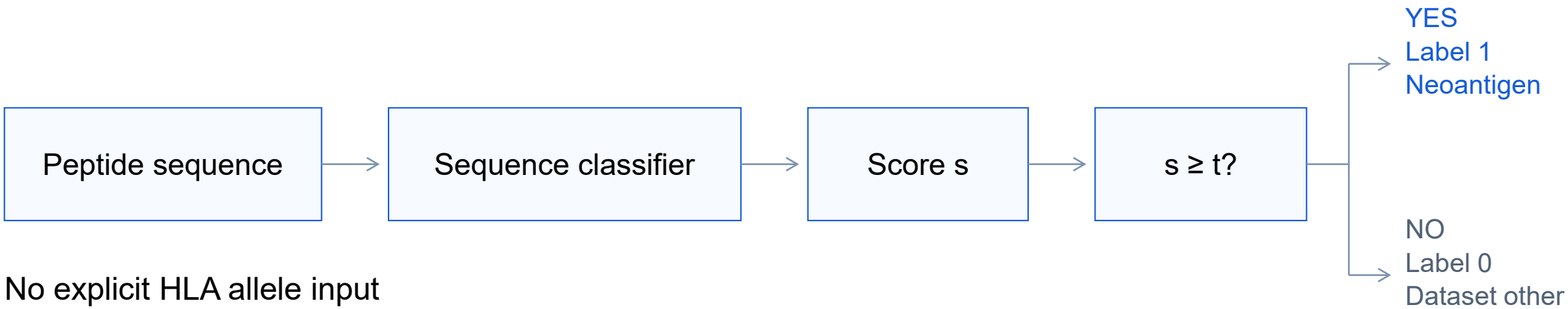
Tumor-specific mutations can alter peptides. Presentation and T-cell recognition depend on biological context.



Biological context only. This classifier does not establish presentation, immune response or vaccine effectiveness.

Defining the prediction problem

A ranking score becomes a binary decision at a locked threshold



No explicit HLA allele input

t selected from training OOF predictions

A score is not proof of immunogenicity or a calibrated probability. Label 0 reflects the dataset definition.

Five questions behind the model sequence

Each approach is trained independently

Approach	Research question
Composition + LR	Are residue frequencies sufficient?
One-hot + MCNN	What can ordered learning from labels achieve?
ESM pooled LR	Do frozen pretrained representations add signal?
ESM CNN	Does local extraction help contextual embeddings?
ESM MCNN	Do multiple local scales help, with added capacity?

Composition versus one-hot changes representation and classifier. Later matched comparisons isolate components more closely.

Dataset provenance and published splits

The independent test keeps its published boundary

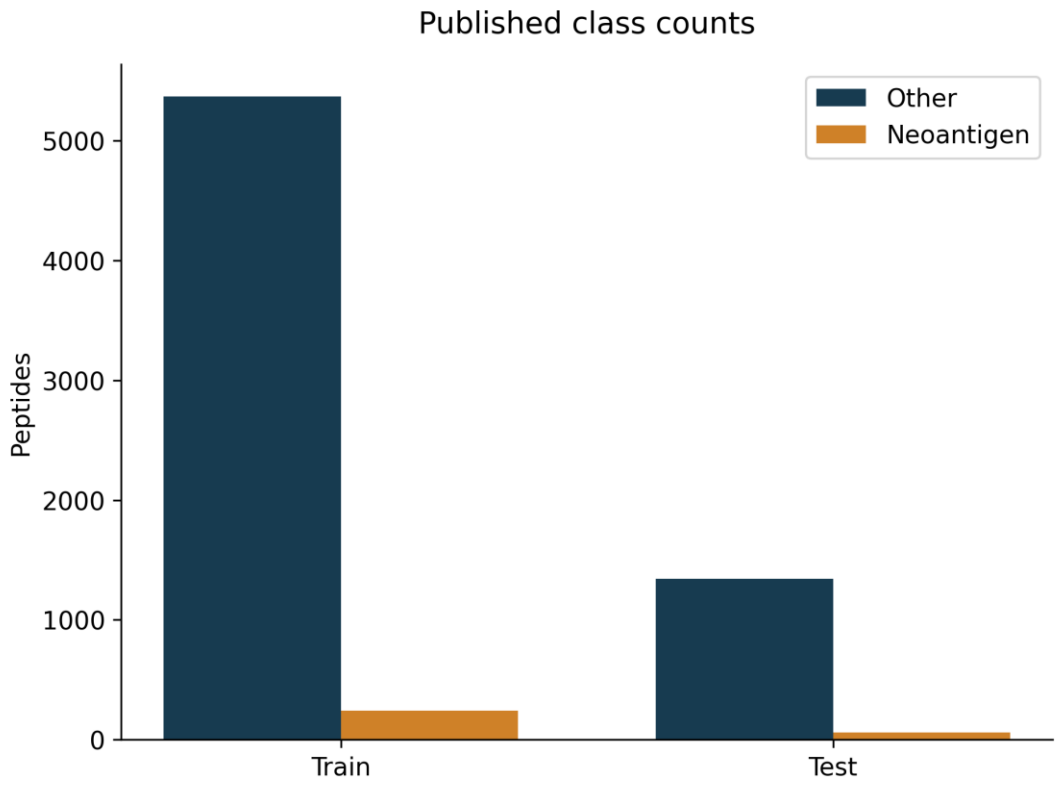


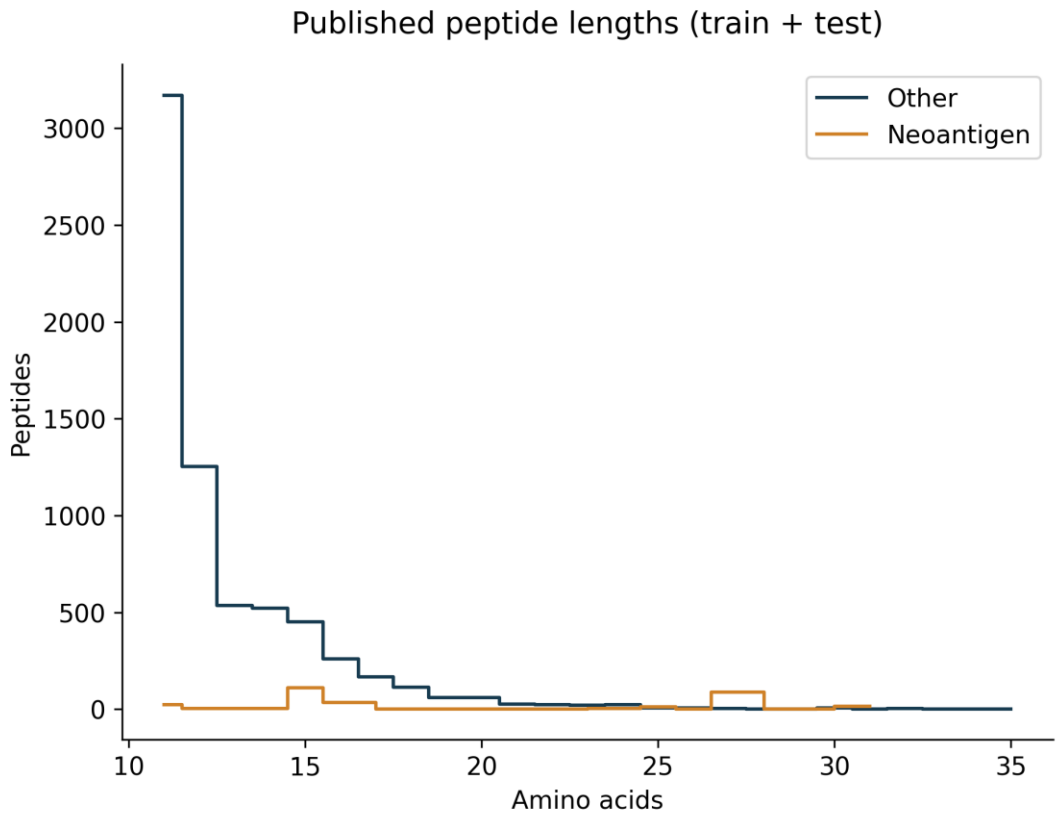
Partition	Neoantigen (1)	Other (0)	Total
Training	241	5,367	5,608
Independent test	61	1,343	1,404
Total	302	6,710	7,012

Record IDs and source labels are retained. Source similarity filtering has not been independently reconstructed.

Understanding the data before training

Rare positives and a length-35 boundary case affect the method

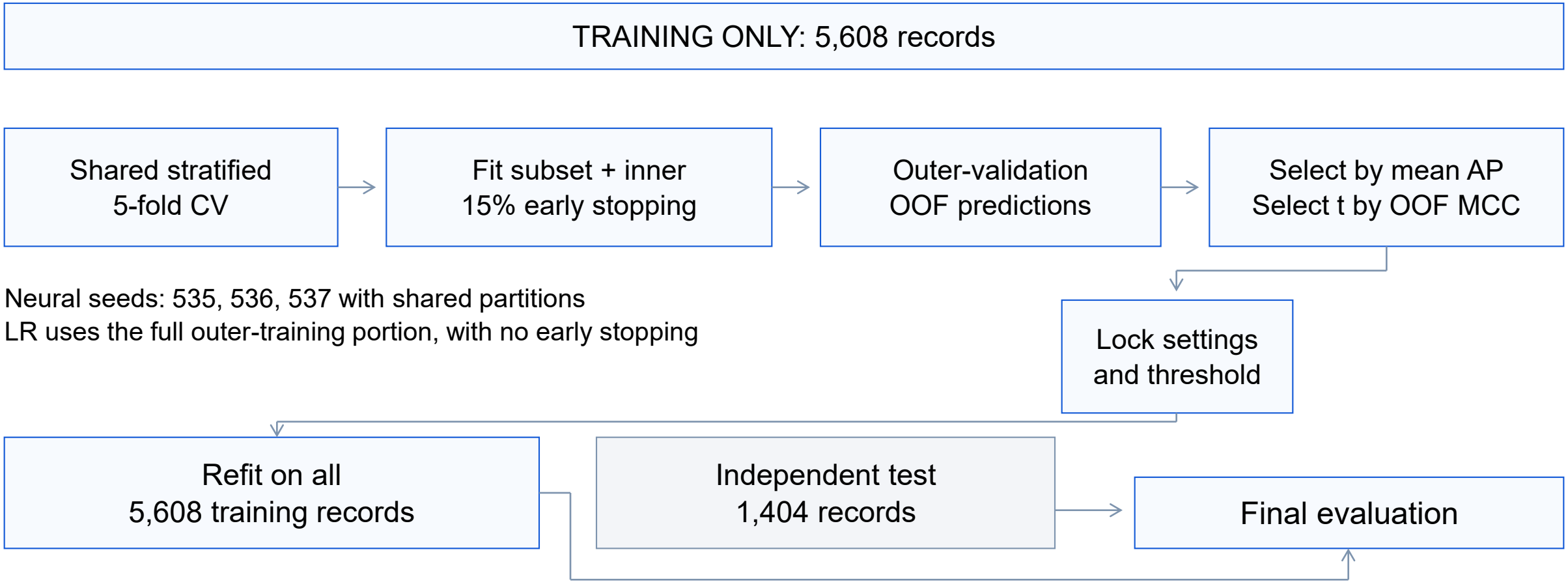




About 1 positive per 22 others. Keep the length-35 record. Length alone does not establish an MHC class.

Designing a fair evaluation

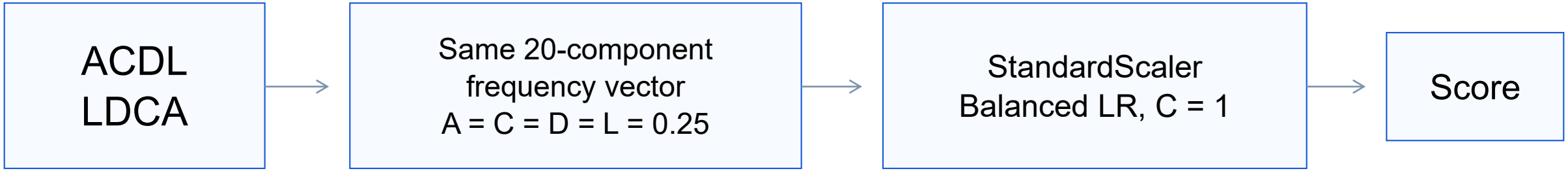
Fitting, early stopping, model selection and testing use distinct roles



Selection CV is optimistic after tuning. Seeds on the same fold are correlated evaluations, not independent datasets.

Baseline 1: amino-acid composition

Composition contains some signal, but loses residue order



Order is lost

Mean CV AP	Test AP	Test MCC	OOF threshold
0.1287	0.0714	0.0515	0.6508

ACDL and LDCA are teaching examples. Test prevalence: 0.0434. The scaler fits within training partitions.

Baseline 2: one-hot MCNN

Ordered local patterns learned directly from task labels

No pretrained encoder

Five parallel windows
k = 2, 3, 4, 5, 8

64 filters per branch
320 concatenated features

Mask padding before max pooling



Trainable parameters	CV AP	Test AP	Test MCC
28,801	0.4943	0.5408	0.5495

Strong efficiency reference. Improvement over composition changes both representation and classifier.

From peptides to contextual embeddings

A frozen ESM encoder gives every real residue a contextual representation

IMAGE PLACEHOLDER
One-hot, embeddings and tensor layout
Prompt C in IMAGE_GENERATION_PROMPTS.md

B
Peptides per batch

L
Padded residue positions

D
Features per residue

8M: D = 320
35M: D = 480

Remove special tokens. Mean pooling uses real residues. CNN windows touching padding are excluded.

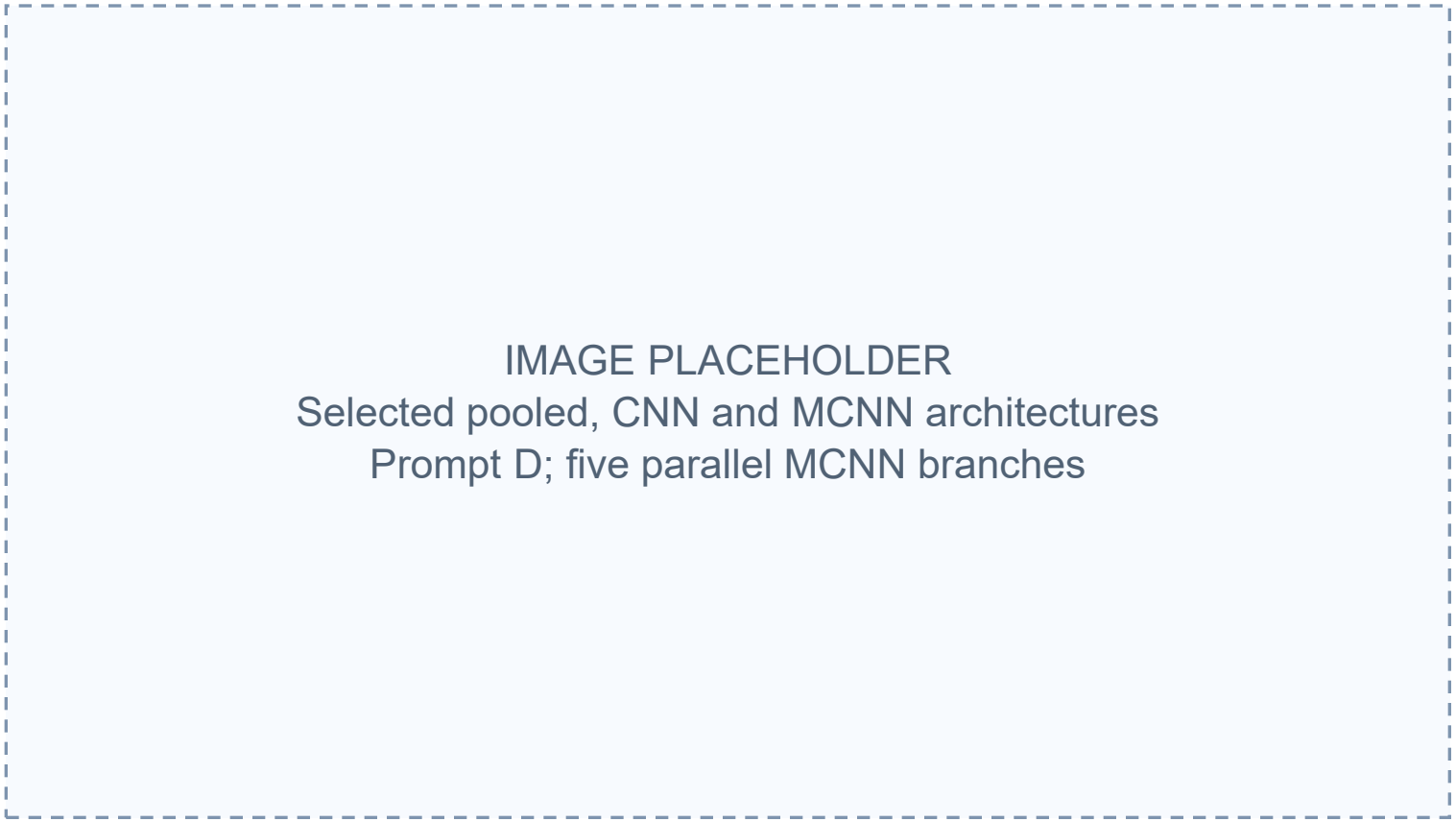
Pooling, single-window CNN and MCNN

The same frozen embeddings support three independent classifiers

Pooled LR
Real-residue mean; C = 0.1
CV / test AP: 0.3976 / 0.4764

Single-window CNN
k = 3; 64 filters
CV / test AP: 0.4868 / 0.5122

Multi-window CNN
256 filters × 5; concat 1,280
CV / test AP: 0.5142 / 0.5437



MCNN adds capacity as well as window diversity. Contextual mean pooling retains information learned by ESM.

Our controlled experiment matrix

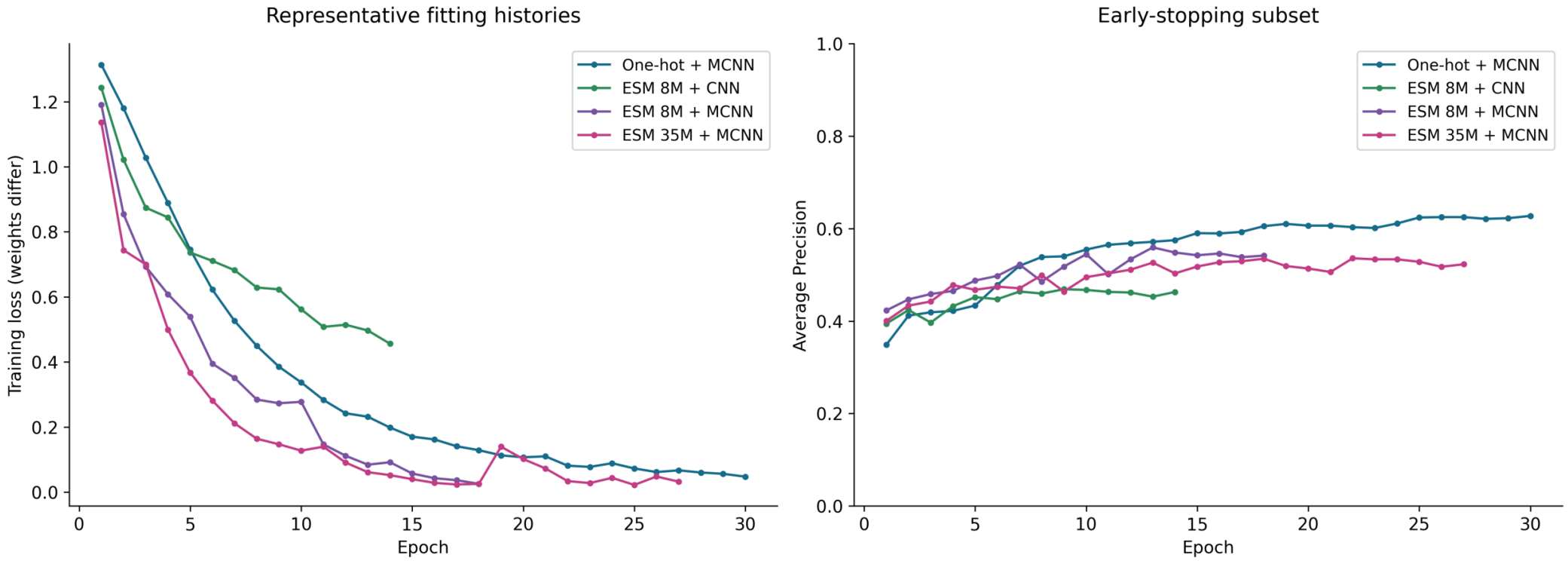
A staged, one-factor search, with 22 configurations

Changed component	Values	Held fixed / caveat
LR regularization	C = 0.1, 1, 10	Balanced LR; shared folds
Single CNN window, 8M	k = 2, 3, 4, 5, 8	64 filters; weighted loss
MCNN filters, 8M	32, 64, 128, 256 per branch	Same five windows; weighted loss
MCNN loss, 8M	Weighted / unweighted	64 filters; same windows
Frozen encoder	8M / 35M	Transfer selected CNN settings; D and head size change

240 CV fits + 18 final refits = 258 fits. AdamW, LR 0.001, weight decay 0.01, batch 64, maximum 30 epochs.

Tracking learning and computational cost

Early-stop AP can plateau while fitting loss continues to decline



Extraction (8M / 35M)

9.53 s / 13.21 s

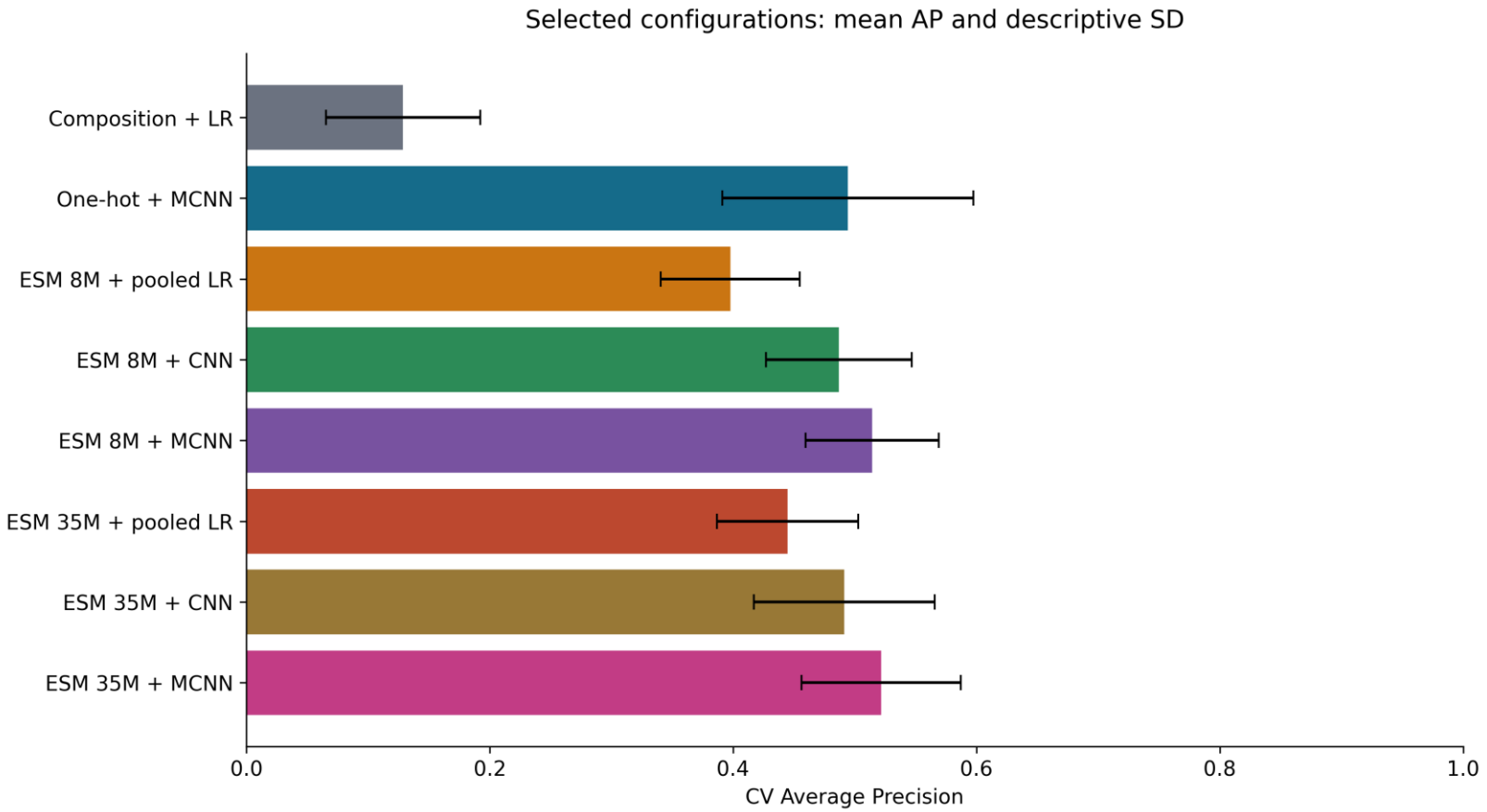
Mean MCNN CV fit (one-hot / 8M / 35M)

10.26 s / 11.50 s / 13.52 s

Fold 0, seed 535. Raw loss scales differ. Timings depend on session state and stopping epoch, not deployment latency.

What we observed across folds and seeds

Mean AP and descriptive spread answer different questions



Mean CV AP ± SD

One-hot MCNN
0.4943 ± 0.1032

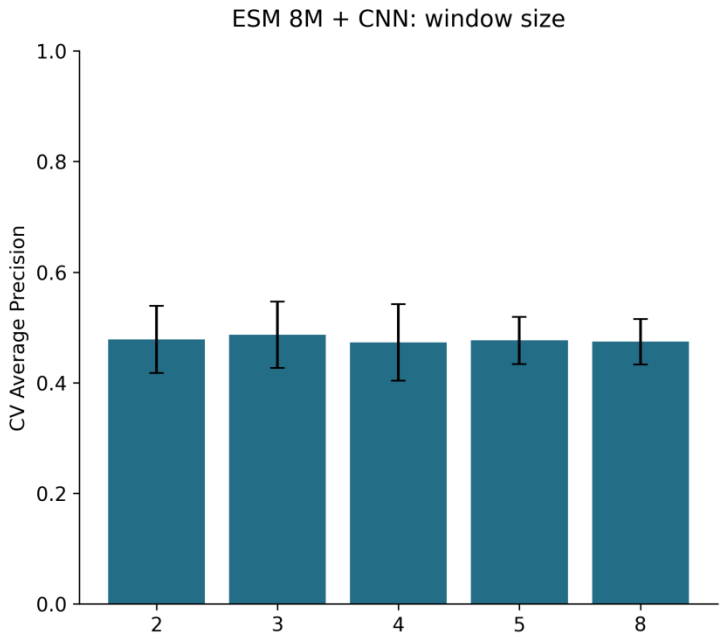
8M MCNN
0.5142 ± 0.0547

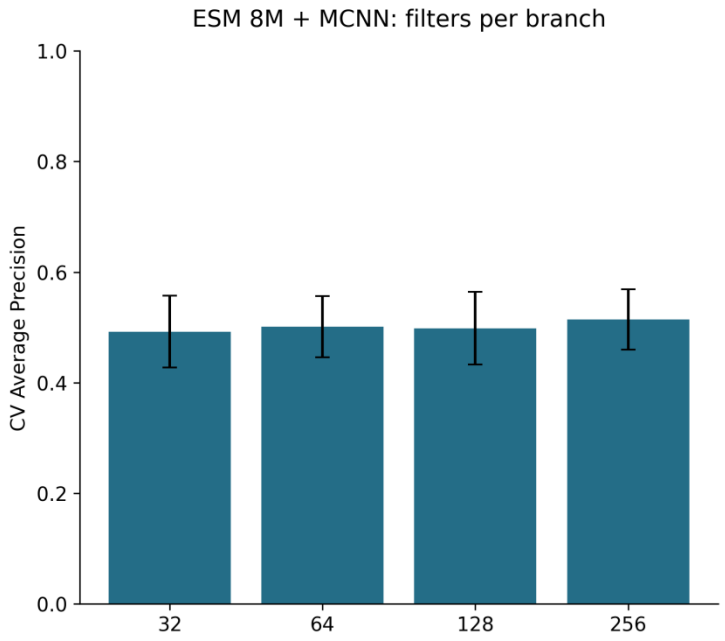
35M MCNN
0.5216 ± 0.0655

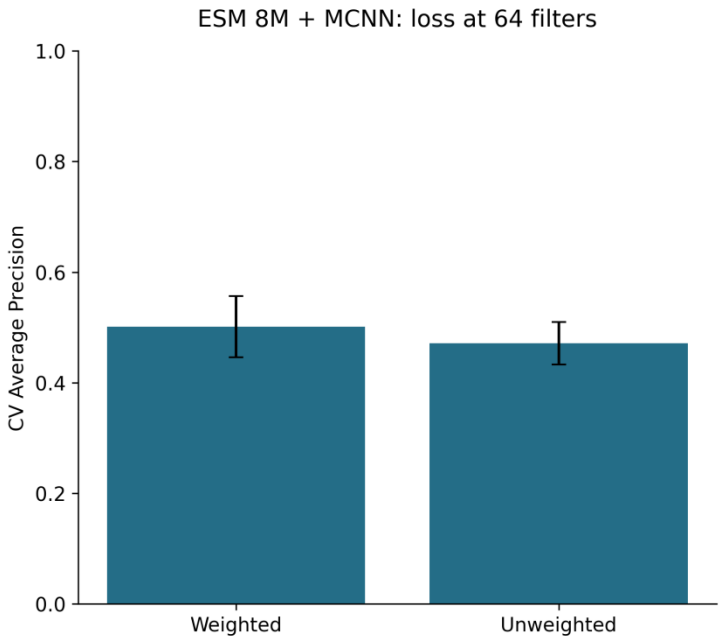
Selection CV, not an unbiased post-selection estimate. Fold 4 is difficult across neural methods; the cause is unresolved.

What changed when one component changed?

Ablations describe conditional effects under the same protocol







Window

k = 3 selected
Best–worst mean AP: 0.0138

Capacity

AP does not rise monotonically
32 / 64 / 128 / 256 filters

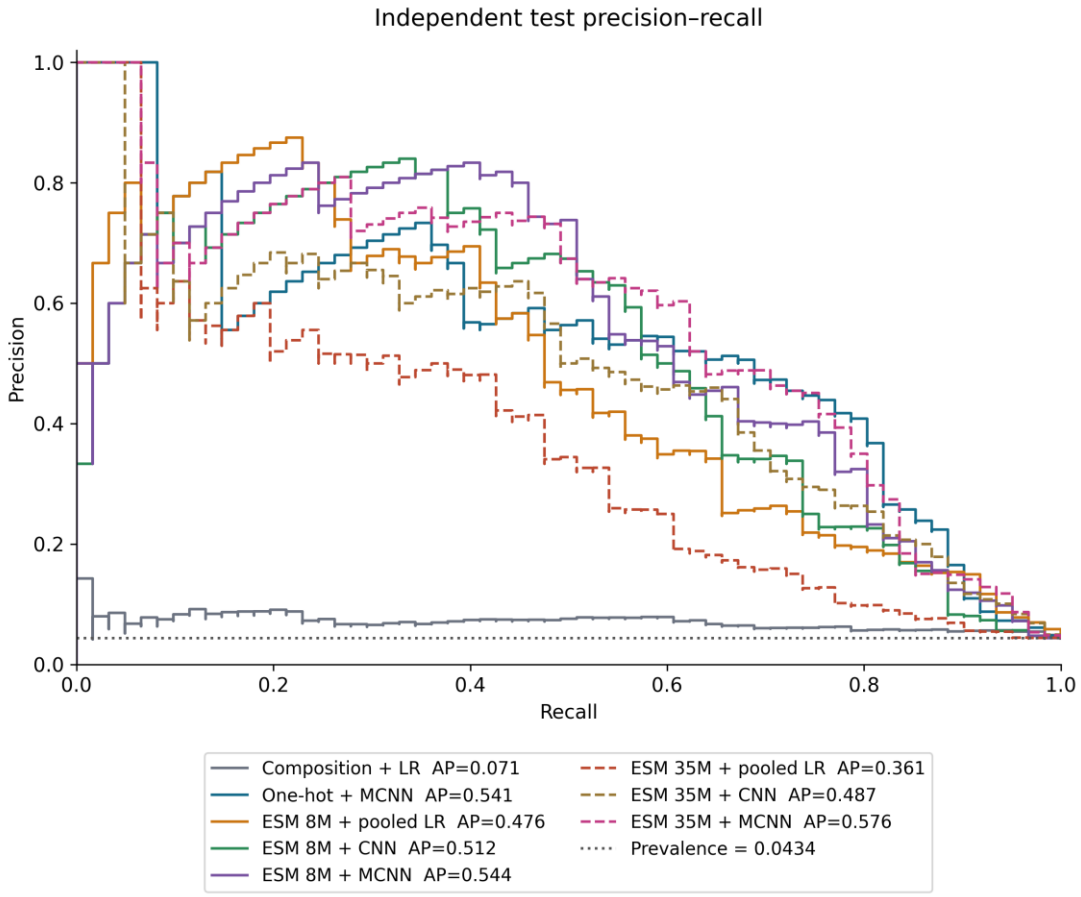
Loss at 64 filters

Weighted AP: 0.5013
Unweighted AP: 0.4713

Error bars show descriptive fold/seed SD. Weighted loss improves AP here, while MCC at 0.5 decreases (0.4225 vs 0.4398).

Independent test: a final check

ESM 35M MCNN has the highest observed AP, with uncertain differences



Locked model	AP	MCC
One-hot MCNN	0.5408	0.5495
8M MCNN	0.5437	0.5512
35M MCNN	0.5758	0.5513

35M MCNN: 44 / 61 positives found
52 false positives

AP 95% CI: [0.4631, 0.7061]

Paired AP difference vs one-hot
95% CI: [−0.0444, 0.1151]

All 1,404 test records. Neural probabilities average three final seeds. Thresholds are locked from OOF MCC.

Errors, limitations and lessons learned

Confident errors require inspection without inventing a biological mechanism

Record ID	Length	Label	Score	Error
sgh_sequence7831	35	0	0.999930	FP
neo_sequence56	11	1	0.000162	FN

ESM 35M + MCNN
Locked threshold: 0.632288

Only 61 positive test records

Source similarity filtering remains unverified

Labels and source domain limit generalization

Sequence appearance does not identify an error mechanism

Examples selected reproducibly by confidence within FP/FN groups. A length-35 error alone does not prove a padding defect.

Conclusions and next experiments

Observed lessons under this dataset and protocol

Question	Observed lesson
Composition sufficient?	Insufficient in this dataset
Ordered supervised patterns?	One-hot MCNN is a strong efficiency reference
Pretrained features?	Useful signal, but pooled classification is limited
Local extraction?	CNN improves over pooled 8M features here
Multiple scales?	Higher observed AP, with added head capacity

Next: similarity-aware validation and an external cohort with locked choices

35M MCNN has the highest observed test AP. Close model differences remain unresolved.

Selected configurations

Appendix A1: settings locked before independent testing

Model	Selected setting	Parameters	OOF t	Refit epochs
Composition + LR	C = 1	21	0.650830	Not applicable
One-hot + MCNN	k=2,3,4,5,8; 64 filters	28,801	0.512465	23
ESM 8M + pooled LR	C = 0.1	321	0.977457	Not applicable
ESM 8M + CNN	k=3; 64 filters	61,569	0.930151	13
ESM 8M + MCNN	k=2,3,4,5,8; 256 filters	1,804,801	0.766193	16
ESM 35M + pooled LR	C = 0.1	481	0.945360	Not applicable
ESM 35M + CNN	k=3; 64 filters	92,289	0.732428	10
ESM 35M + MCNN	k=2,3,4,5,8; 256 filters	2,705,921	0.632288	14

ESM is frozen. Parameters count only the trainable classifier. Neural final predictions average seeds 535, 536, 537.

Complete CV configurations (1/2)

Appendix A2: all 22 settings, including unselected configurations

Model	Setting	Mean AP	SD AP	MCC at 0.5	Params
Composition + LR	C=0.1	0.1286	0.0631	0.1507	21
Composition + LR	C=1	0.1287	0.0635	0.1505	21
Composition + LR	C=10	0.1287	0.0635	0.1505	21
One-hot + MCNN	k=2,3,4,5,8; F=64; weighted	0.4943	0.1032	0.4182	28,801
ESM 8M + pooled LR	C=0.1	0.3976	0.0572	0.3191	321
ESM 8M + pooled LR	C=1	0.3264	0.0431	0.3173	321
ESM 8M + pooled LR	C=10	0.2895	0.0397	0.2853	321
ESM 8M + CNN	k=2; F=64; weighted	0.4782	0.0607	0.3439	41,089
ESM 8M + CNN	k=3; F=64; weighted	0.4868	0.0600	0.3603	61,569
ESM 8M + CNN	k=4; F=64; weighted	0.4730	0.0692	0.3668	82,049
ESM 8M + CNN	k=5; F=64; weighted	0.4765	0.0427	0.3560	102,529

Mean and sample SD over saved outer-fold/seed rows. Selection reuses CV, so these are optimistic after tuning.

Complete CV configurations (2/2)

Appendix A3: all 22 settings, including unselected configurations

Model	Setting	Mean AP	SD AP	MCC at 0.5	Params
ESM 8M + CNN	k=8; F=64; weighted	0.4740	0.0413	0.3604	163,969
ESM 8M + MCNN	k=2,3,4,5,8; F=32; weighted	0.4922	0.0651	0.3718	225,601
ESM 8M + MCNN	k=2,3,4,5,8; F=64; weighted	0.5013	0.0551	0.4225	451,201
ESM 8M + MCNN	k=2,3,4,5,8; F=128; weighted	0.4985	0.0659	0.4280	902,401
ESM 8M + MCNN	k=2,3,4,5,8; F=256; weighted	0.5142	0.0547	0.4594	1,804,801
ESM 8M + MCNN	k=2,3,4,5,8; F=64; unweighted	0.4713	0.0384	0.4398	451,201
ESM 35M + pooled LR	C=0.1	0.4449	0.0581	0.3712	481
ESM 35M + pooled LR	C=1	0.3812	0.0319	0.3714	481
ESM 35M + pooled LR	C=10	0.3193	0.0267	0.3631	481
ESM 35M + CNN	k=3; F=64; weighted	0.4913	0.0744	0.3769	92,289
ESM 35M + MCNN	k=2,3,4,5,8; F=256; weighted	0.5216	0.0655	0.4645	2,705,921

Mean and sample SD over saved outer-fold/seed rows. Selection reuses CV, so these are optimistic after tuning.

Fixed defaults versus tuned selections

Appendix A4: comparison includes the model-selection step

Model	Default CV AP	Selected CV AP	Default params	Selected params
Composition + LR	0.1287	0.1287	21	21
One-hot + MCNN	0.4943	0.4943	28,801	28,801
ESM 8M + pooled LR	0.3264	0.3976	321	321
ESM 8M + CNN	0.4868	0.4868	61,569	61,569
ESM 8M + MCNN	0.5013	0.5142	451,201	1,804,801

Defaults and selected settings are not independent estimates. A larger selected head changes capacity as well as window coverage.

Metric definitions and threshold policy

Appendix A5: ranking quality and a binary operating point answer different questions

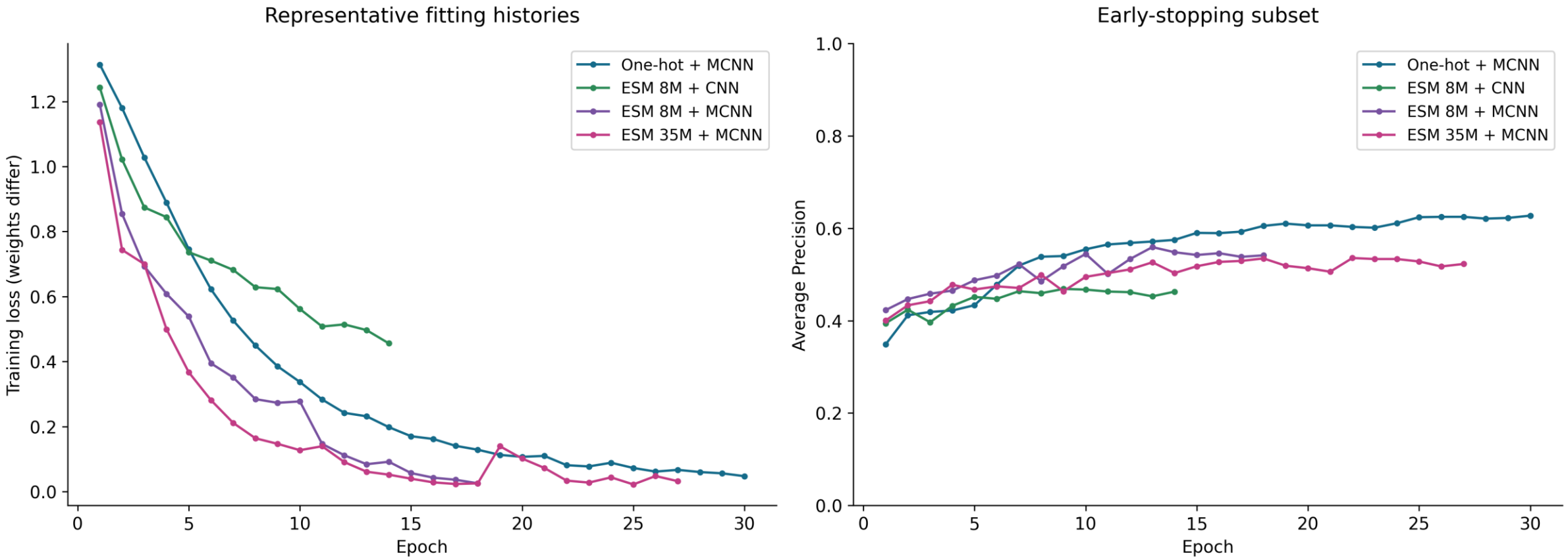
Measure	Meaning
Average precision (AP)	Summarizes precision across recall changes in a ranked list
ROC-AUC	Ranking discrimination across false-positive-rate levels
Precision	$TP / (TP + FP)$
Recall / sensitivity	$TP / (TP + FN)$
Specificity	$TN / (TN + FP)$
MCC	$(TP \times TN - FP \times FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$

Configuration: maximize mean CV AP. Threshold: maximize MCC from seed-averaged training OOF predictions. Independent test uses both 0.5 and the locked OOF threshold. Main slides use the OOF threshold.

AP and ROC-AUC use scores. Precision, recall, specificity, MCC and confusion counts depend on the threshold.

Complete learning histories

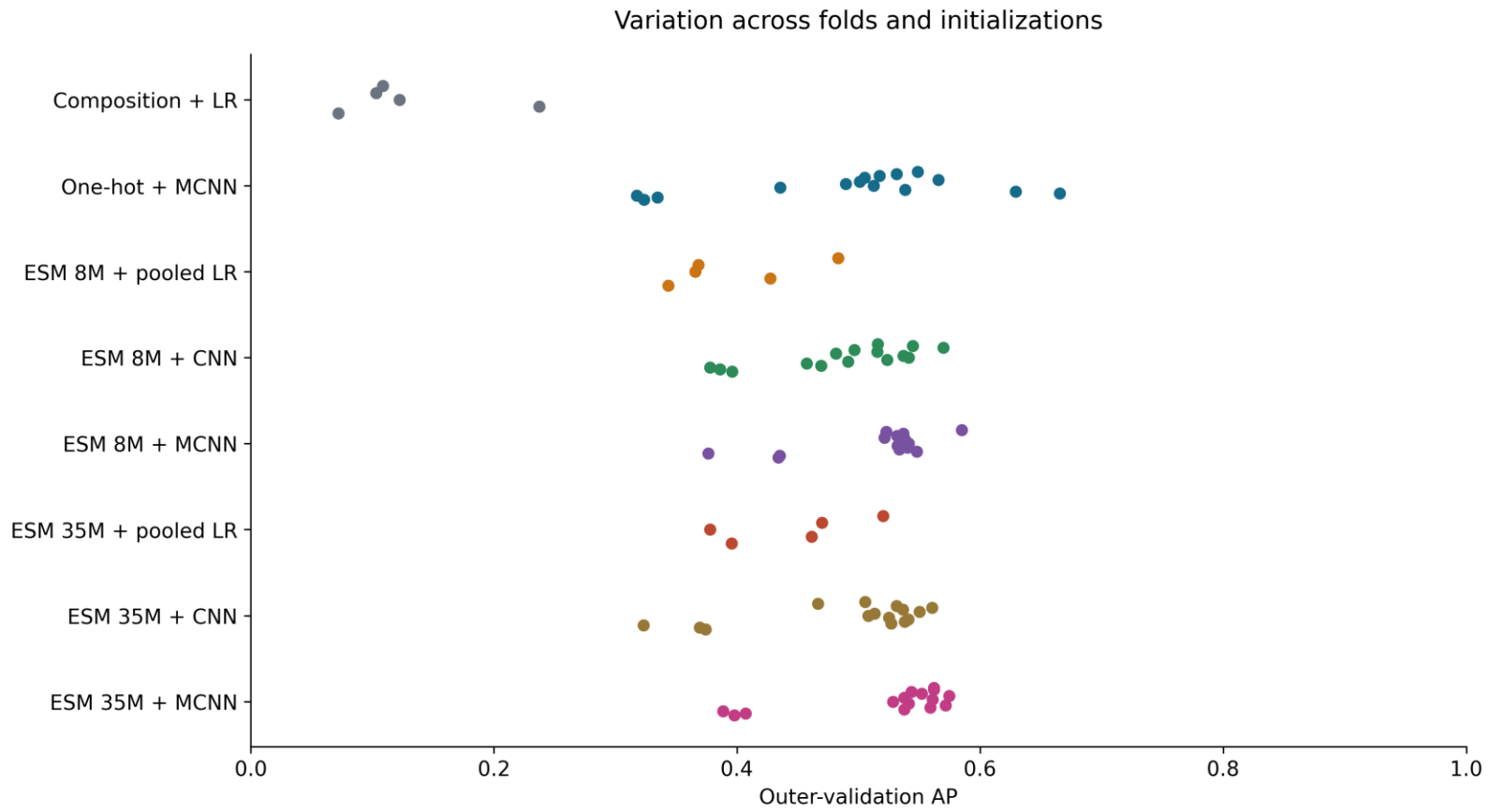
Appendix A6: selected settings, first fold and first seed



Fold 0, seed 535. Raw weighted losses have different scales. Early-stop AP supports checkpoint selection.

Fold and seed variation

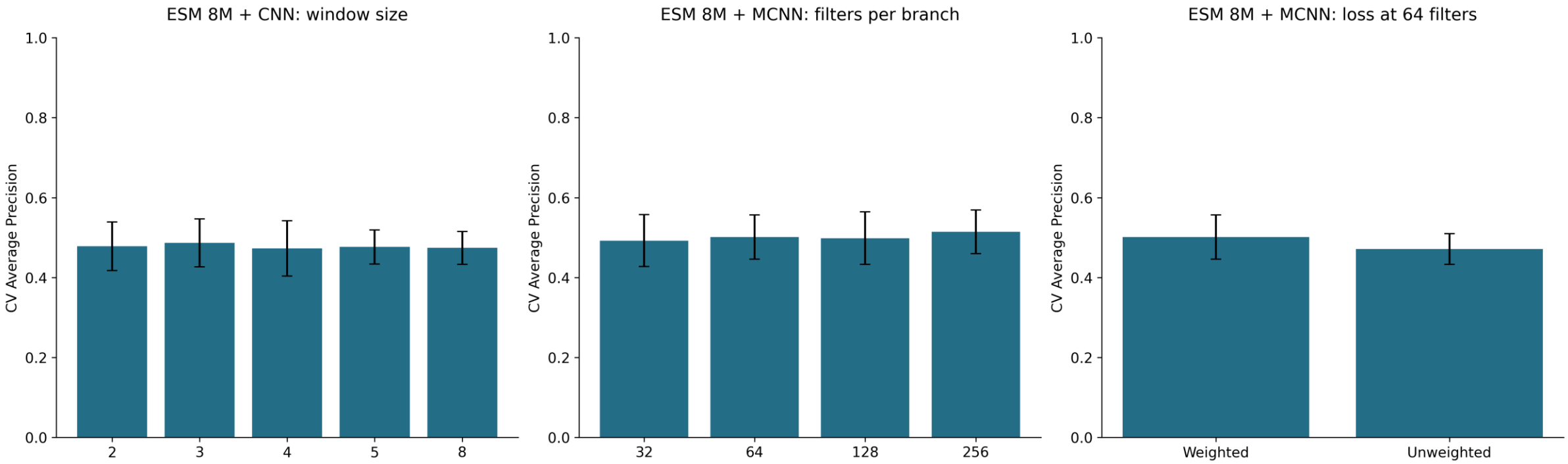
Appendix A7: each point is one run on the same published training split



Seeds and folds are correlated. Spread is descriptive, not a confidence interval. Difficult fold causes remain unresolved.

Complete controlled ablations

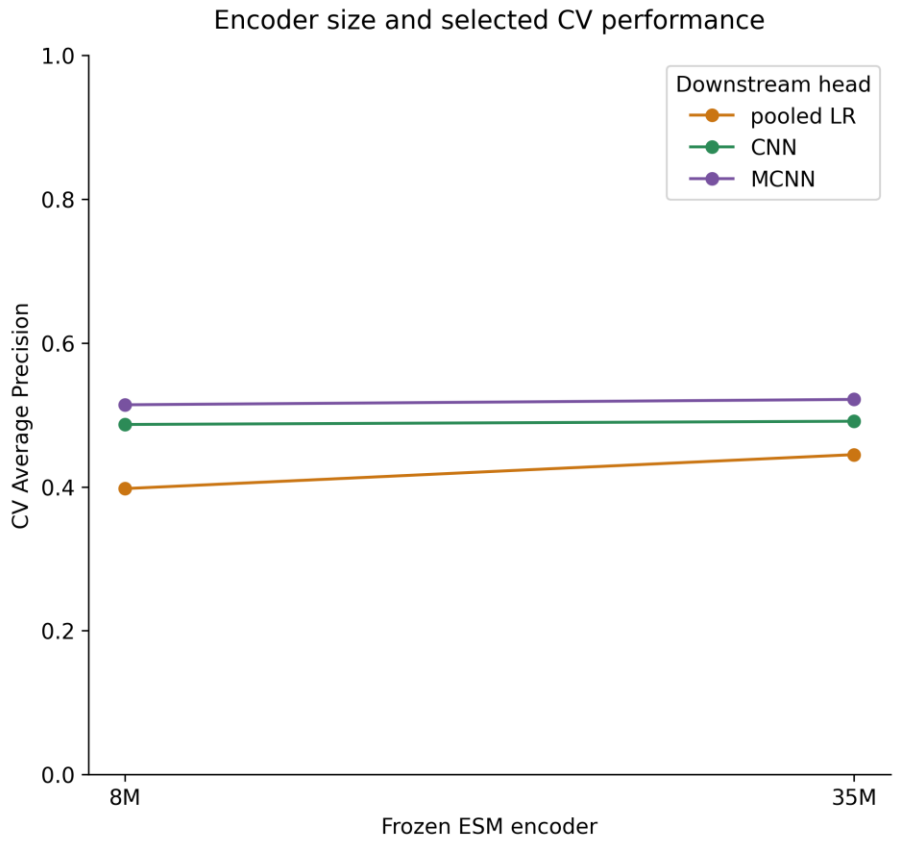
Appendix A8: one-factor 8M comparisons

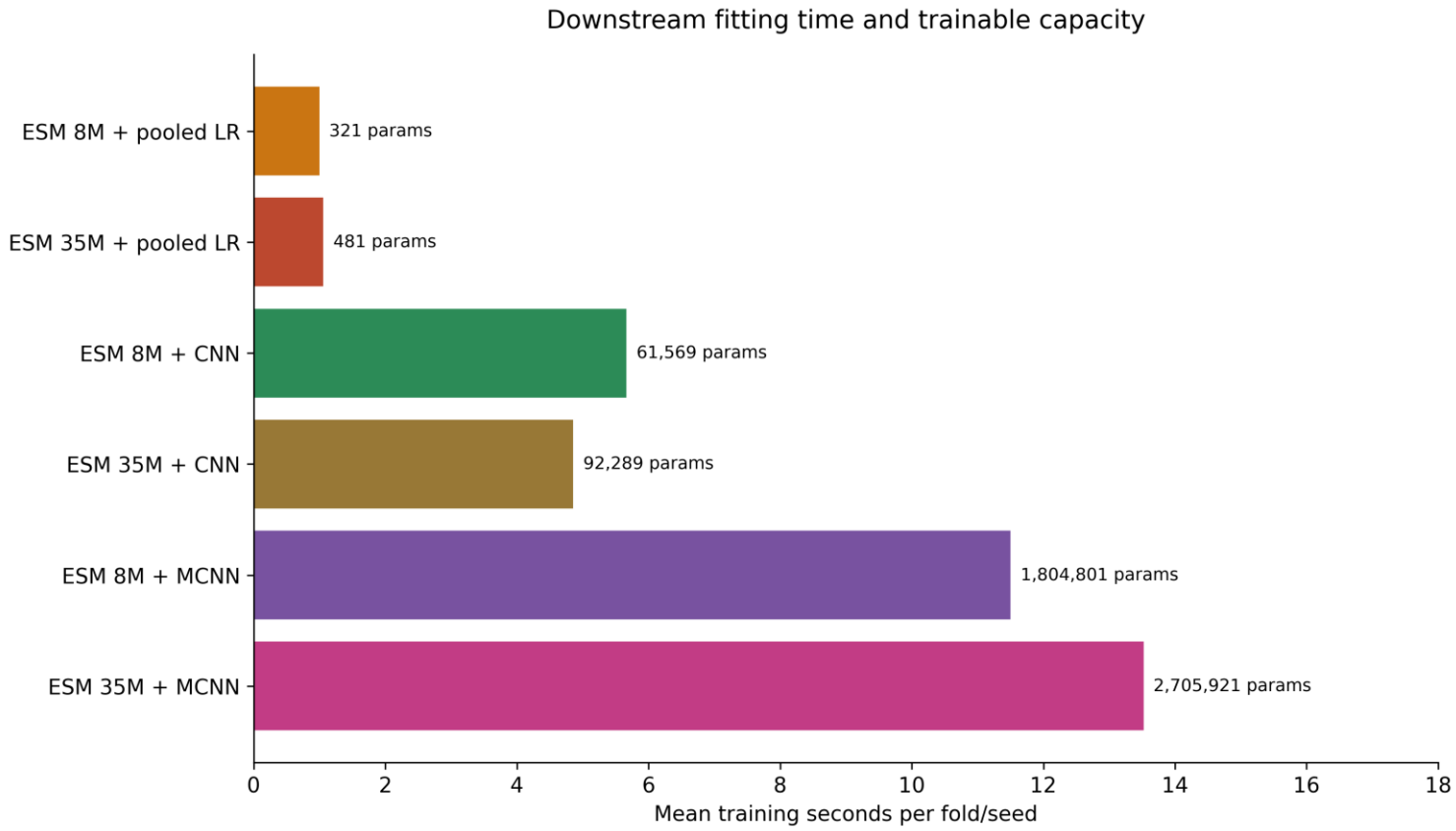


Error bars show descriptive fold/seed SD. Window, filter and loss contrasts retain their fixed conditions.

Encoder size, head quality and cost

Appendix A9: transferred CNN settings, with changed D and head parameter count





8M / 35M change embedding dimension and head capacity. Downstream fit time excludes extraction and depends on hardware.

Complete independent-test metrics

Appendix A10: ensemble probabilities and locked OOF thresholds

Model	AP	ROC-AUC	MCC	Precision	Recall	Specificity	TP / FP
Composition + LR	0.0714	0.6625	0.0515	0.0690	0.2295	0.8593	14 / 189
One-hot + MCNN	0.5408	0.9179	0.5495	0.5362	0.6066	0.9762	37 / 32
ESM 8M + pooled LR	0.4764	0.9077	0.4743	0.6667	0.3607	0.9918	22 / 11
ESM 8M + CNN	0.5122	0.8880	0.5262	0.7143	0.4098	0.9926	25 / 10
ESM 8M + MCNN	0.5437	0.9108	0.5512	0.6000	0.5410	0.9836	33 / 22
ESM 35M + pooled LR	0.3614	0.8213	0.3693	0.5143	0.2951	0.9873	18 / 17
ESM 35M + CNN	0.4870	0.9154	0.4938	0.4557	0.5902	0.9680	36 / 43
ESM 35M + MCNN	0.5758	0.9224	0.5513	0.4583	0.7213	0.9613	44 / 52

N = 1,404, including 61 positives and 1,343 others. LR has one refit. Neural probabilities average three refits.

Test uncertainty and seed spread

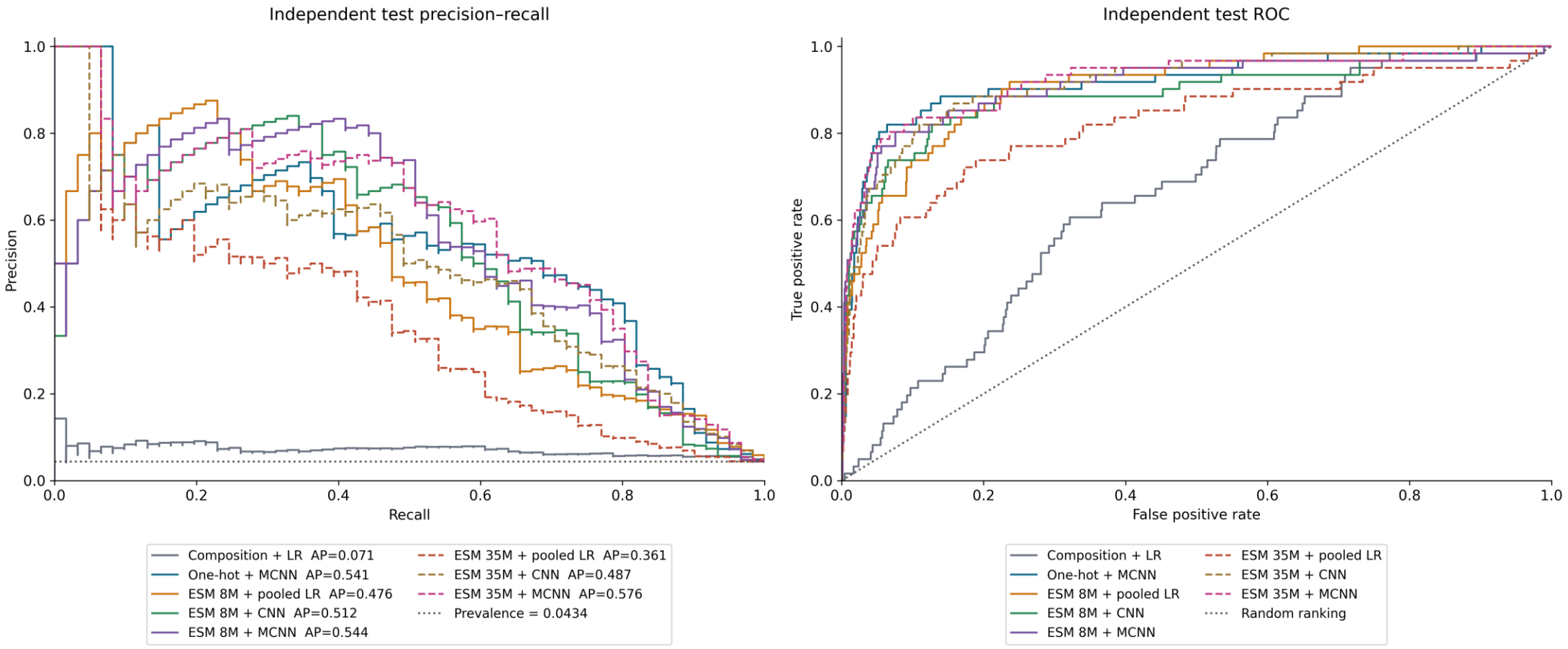
Appendix A11: 2,000 stratified bootstrap repetitions, fixed trained models

Model	Test AP	AP 95% interval	MCC 95% interval	Single-seed AP range
Composition + LR	0.0714	[0.0577, 0.1057]	[-0.0052, 0.1140]	0.0714–0.0714
One-hot + MCNN	0.5408	[0.4347, 0.6662]	[0.4465, 0.6465]	0.5112–0.5373
ESM 8M + pooled LR	0.4764	[0.3655, 0.6187]	[0.3517, 0.5934]	0.4764–0.4764
ESM 8M + CNN	0.5122	[0.4025, 0.6606]	[0.4026, 0.6394]	0.5094–0.5100
ESM 8M + MCNN	0.5437	[0.4322, 0.6933]	[0.4479, 0.6544]	0.5211–0.5328
ESM 35M + pooled LR	0.3614	[0.2573, 0.4934]	[0.2484, 0.4912]	0.3614–0.3614
ESM 35M + CNN	0.4870	[0.3786, 0.6198]	[0.3998, 0.5905]	0.4667–0.4966
ESM 35M + MCNN	0.5758	[0.4631, 0.7061]	[0.4681, 0.6338]	0.5696–0.5833

Exploratory paired AP difference, 35M MCNN minus one-hot MCNN: 0.0350, 95% CI [−0.0444, 0.1151].
No multiplicity correction. These intervals do not cover retraining or source-domain uncertainty.

Complete test precision–recall and ROC

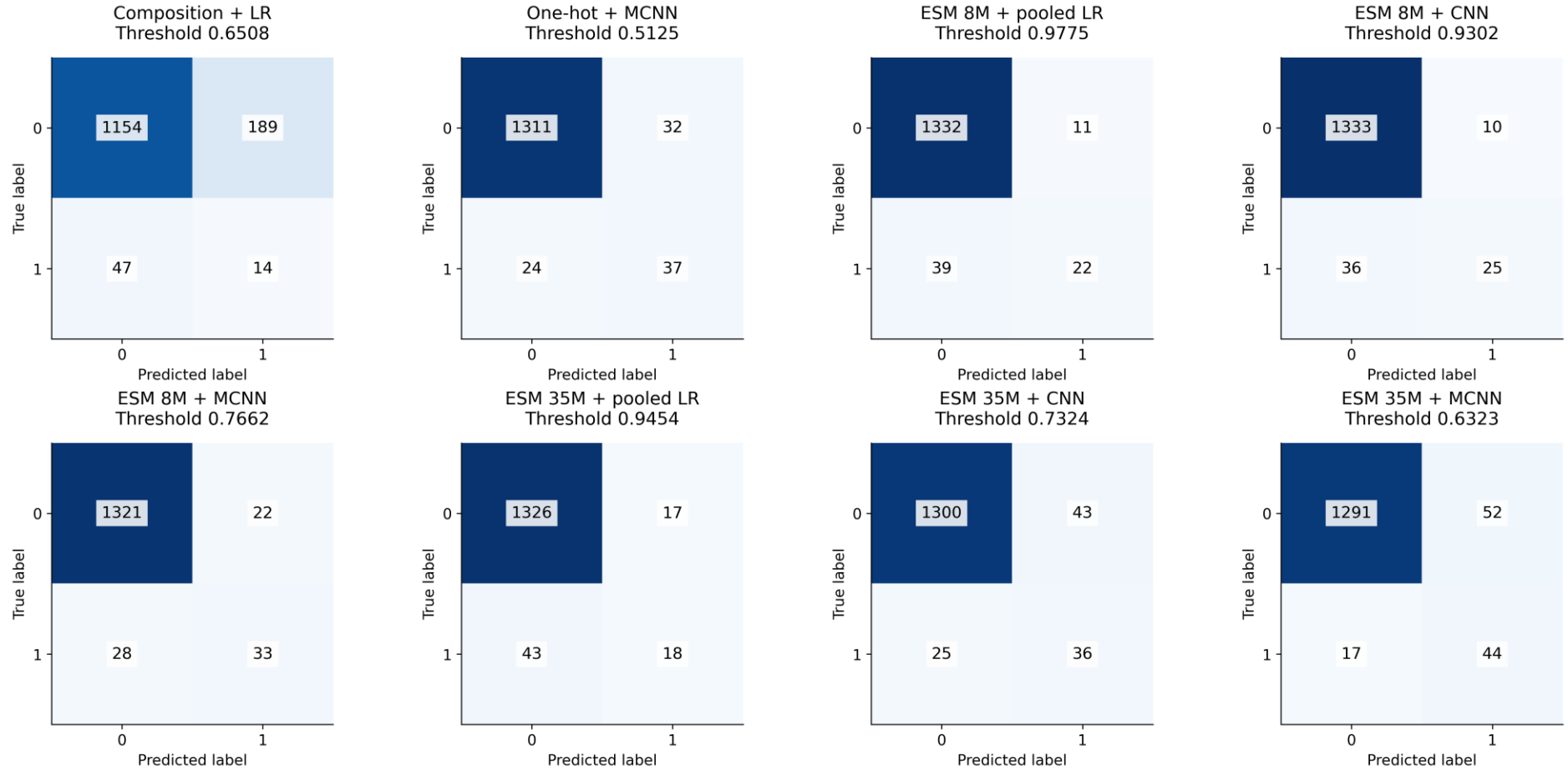
Appendix A12: all eight locked methods on the same independent test



N = 1,404. PR prevalence = 0.0434. Neural scores average three seed refits. No test-driven configuration changes.

All eight confusion matrices

Appendix A13: raw counts at locked OOF-selected thresholds



Label 1: neoantigen. Label 0: dataset other. Common color scale. Every panel covers the same 1,404 records.

Additional representative errors

Appendix A14: two FP and two FN examples for selected multi-window models

Model	Record	Length	Label	Score	Threshold	Error
One-hot + MCNN	neo_sequence56	11	1	0.000960	0.512465	FN
One-hot + MCNN	neo_sequence664	15	1	0.003104	0.512465	FN
One-hot + MCNN	sgh_sequence28676	30	0	0.964475	0.512465	FP
One-hot + MCNN	sgh_sequence19933	16	0	0.962445	0.512465	FP
ESM 35M + MCNN	neo_sequence56	11	1	0.000162	0.632288	FN
ESM 35M + MCNN	neo_sequence664	15	1	0.000451	0.632288	FN
ESM 35M + MCNN	sgh_sequence7831	35	0	0.999930	0.632288	FP
ESM 35M + MCNN	sgh_sequence7861	26	0	0.999847	0.632288	FP

Deterministic confidence-based examples, not a mechanistic explanation. Full sequences and all model examples remain in the source CSV.

Execution provenance and limitations

Appendix A15: numerical reconciliation does not establish exact execution-source identity

258 completed fits and saved numerical evidence reconcile

The recorded run hash differs from the current pipeline and notebook. The exact executed notebook snapshot is missing.

Recorded full-run code SHA-256
3bd6cb69a8bfeca0a63bf91c461152c45312a7e5c3b9ff55308d7090009b997e

Current scripts/course_pipeline.py SHA-256
a01bdafcf50131f31f1cc5bca963e66cfa6f9a6ac4240bbada93a6ec33fcdf95

Next evidence: archive the executed notebook, audit sequence similarity reproducibly, then evaluate locked choices on an external cohort.

Source-domain generalization, label quality and only 61 positive test cases limit the interpretation.

References and image credits

Appendix A16: sources for biological context, methods and empirical figures

Chuang et al. (2024), International Journal of Biological Macromolecules
https://doi.org/10.1016/j.ijbiomac.2024.136252
Dataset and original project: https://github.com/B1607/DeepNeoAG

ESM models and implementation
https://github.com/facebookresearch/esm

Slide 2 illustration: adapted from Nature (2023) via National Cancer Institute
doi:10.1038/s41586-023-06063-y, CC BY 4.0. Resized only.
https://creativecommons.org/licenses/by/4.0/

Empirical figures: complete supplied training run, results_full/
Named figure exports: docs/assets/full_results/
Operating Review template: retained source deck, cloned and adapted

No AI-generated illustration is presented. Conceptual architecture placeholders remain for later image insertion.